Top 100 Orthopedic AI • Rank #48

How to Develop and Validate Prediction Models for Orthopedic Outcomes

I. Zaniletti, D. Larson, D. Lewallen, D. Berry, H. Maradit Kremers

Department of Quantitative Health Sciences, Mayo Clinic, Scottsdale, Arizona.

Journal:	Publication Year:	Citations:	DOI:
The Journal of arthroplasty	2022.0	13	10.1016/j.arth.2022.12.032

CLINICAL & AGENTIC AI TAKEAWAY

Prediction models can be valuable tools for supporting surgical decision-making in arthroplasty, but require robust construction and validation for widespread adoption in clinical practice.

ABSTRACT

--- REPOSITORY & PUBLISHER WEB METADATA --- Affiliations: Department of Quantitative Health Sciences, Mayo Clinic, Scottsdale, Arizona. Keywords: Model validation, Orthopedics, Arthroplasty, Predictors, Machine Learning, Risk Prediction Extended Abstract from Repository: Prediction models are common in medicine for predicting outcomes such as mortality, complications, or response to treatment. Despite the growing interest in these models in arthroplasty (and orthopaedics in general), few have been adopted in clinical practice. If robustly built and validated, prediction models can be excellent tools to support surgical decision making. In this paper, we provide an overview of the statistical concepts surrounding prediction models and outline practical steps for prediction model development and validation in arthroplasty research. Please visit the followinghttps://www.youtube.com/watch?v=9Yrit23Rkicfor a video that explains the highlights of the paper in practical terms.

Keywords: Model validation, Orthopedics, Arthroplasty, Predictors, Machine Learning, Risk Prediction

Study Overview & Background

--- REPOSITORY & PUBLISHER WEB METADATA --- Affiliations: Department of Quantitative Health Sciences, Mayo Clinic, Scottsdale, Arizona. Keywords: Model validation, Orthopedics, Arthroplasty, Predictors, Machine Learning, Risk Prediction Extended Abstract from Repository: Prediction models are common in medicine for predicting outcomes such as mortality, complications, or response to treatment. Despite the growing interest in these models in arthroplasty (and orthopaedics in general), few have been adopted in clinical practice. If robustly built and validated, prediction models can be excellent tools to support surgical decision making. In this paper, we provide an overview of the statistical concepts surrounding prediction models and outline practical steps for prediction model development and validation in arthroplasty research. Please visit the followinghttps://www.youtube.com/watch?v=9Yrit23Rkicfor a video that explains the highlights of the paper in practical terms.

Methodology & Clinical Focus

Study Classification: other. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

Prediction models can be valuable tools for supporting surgical decision-making in arthroplasty, but require robust construction and validation for widespread adoption in clinical practice.

Extended Repository Abstract

Prediction models are common in medicine for predicting outcomes such as mortality, complications, or response to treatment. Despite the growing interest in these models in arthroplasty (and orthopaedics in general), few have been adopted in clinical practice. If robustly built and validated, prediction models can be excellent tools to support surgical decision making. In this paper, we provide an overview of the statistical concepts surrounding prediction models and outline practical steps for prediction model development and validation in arthroplasty research. Please visit the followinghttps://www.youtube.com/watch?v=9Yrit23Rkicfor a video that explains the highlights of the paper in practical terms.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

- **Authors:** I. Zaniletti, D. Larson, D. Lewallen, D. Berry, H. Maradit Kremers
- **Journal / Venue:** The Journal of arthroplasty (2022.0)
- **SJR Metric:** Quartile 1.0
- **Citation Count:** 13 citations recorded
- **Digital Object Identifier (DOI):** https://doi.org/10.1016/j.arth.2022.12.032
- **Consensus Evidence Link:** https://consensus.app/papers/how-to-develop-and-validate-prediction-models-for-zaniletti-larson/82d328c8719c59f5a73aa44dde1b9415/

Source & Provenance: Curated for Agentic Orthopedics Research Collection • DOI: https://doi.org/10.1016/j.arth.2022.12.032 • Consensus Page: https://consensus.app/papers/how-to-develop-and-validate-prediction-models-for-zaniletti-larson/82d328c8719c59f5a73aa44dde1b9415/ • Verified and rendered via Deep Search & Academic Repositories.